

Program 1: VHDL Code for Full Adder

```
G:/code/vhdl/full_adder.vhd - Default
Ln#
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity full_adder is
5  Port(
6      A:in std_logic;
7      B:in std_logic;
8      Cin:in std_logic;
9      Sum:out std_logic;
10     Cout:out std_logic
11 );
12 end full_adder;
13
14 architecture Behavioral of full_adder is
15 begin
16     Sum <= A xor B xor Cin;
17     Cout <= (A and B) or (Cin and (A xor B));
18 end Behavioral;
```

Program 2: VHDL Code for 16-bit Ripple Carry Adder (RCA)

```
G:/code/vhdl/rca_16bit.vhd - Default
Ln#
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity rca_16bit is
5  Port(
6      A:in std_logic_vector(15 downto 0);
7      B:in std_logic_vector(15 downto 0);
8      Cin:in std_logic;
9      Sum:out std_logic_vector(15 downto 0);
10     Cout:out std_logic
11 );
12 end rca_16bit;
13
14 architecture structural of rca_16bit is
15 component full_adder
16 Port(
17     A:in std_logic;
18     B:in std_logic;
19     Cin:in std_logic;
20     Sum:out std_logic;
21     Cout:out std_logic
22 );
23 end component;
24 signal C: std_logic_vector(16 downto 0);
25 begin
26     FA0: full_adder port map(A(0),B(0),Cin,Sum(0),C(0));
27     FA1: full_adder port map(A(1),B(1),C(0),Sum(1),C(1));
28     FA2: full_adder port map(A(2),B(2),C(1),Sum(2),C(2));
29     FA3: full_adder port map(A(3),B(3),C(2),Sum(3),C(3));
30     FA4: full_adder port map(A(4),B(4),C(3),Sum(4),C(4));
31     FA5: full_adder port map(A(5),B(5),C(4),Sum(5),C(5));
32     FA6: full_adder port map(A(6),B(6),C(5),Sum(6),C(6));
33     FA7: full_adder port map(A(7),B(7),C(6),Sum(7),C(7));
34     FA8: full_adder port map(A(8),B(8),C(7),Sum(8),C(8));
35     FA9: full_adder port map(A(9),B(9),C(8),Sum(9),C(9));
36     FA10: full_adder port map(A(10),B(10),C(9),Sum(10),C(10));
37     FA11: full_adder port map(A(11),B(11),C(10),Sum(11),C(11));
38     FA12: full_adder port map(A(12),B(12),C(11),Sum(12),C(12));
39     FA13: full_adder port map(A(13),B(13),C(12),Sum(13),C(13));
40     FA14: full_adder port map(A(14),B(14),C(13),Sum(14),C(14));
41     FA15: full_adder port map(A(15),B(15),C(14),Sum(15),Cout);
42
43 end structural;
```

Program 3: VHDL Code for 16-bit ALU

```
G:/code/vhdl/alu_16bit.vhd (/tb_alu_16bit/UUT) - Default
Ln#
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity alu_16bit is
5  Port (
6      A      : in  STD_LOGIC_VECTOR(15 downto 0);
7      B      : in  STD_LOGIC_VECTOR(15 downto 0);
8      sel    : in  STD_LOGIC_VECTOR(1 downto 0);
9      Result : out STD_LOGIC_VECTOR(15 downto 0);
10     Cout   : out STD_LOGIC
11 );
12 end alu_16bit;
13
14 architecture Structural of alu_16bit is
15
16     component rca_16bit
17     Port (
18         A      : in  STD_LOGIC_VECTOR(15 downto 0);
19         B      : in  STD_LOGIC_VECTOR(15 downto 0);
20         Cin    : in  STD_LOGIC;
21         Sum    : out STD_LOGIC_VECTOR(15 downto 0);
22         Cout   : out STD_LOGIC
23     );
24 end component;
25
26 signal B_modified : STD_LOGIC_VECTOR(15 downto 0);
27 signal Add_Result : STD_LOGIC_VECTOR(15 downto 0);
28 signal Add_Cout   : STD_LOGIC;
29 signal Add_Cin    : STD_LOGIC;
30
31 begin
32
33     -- For subtraction, complement B
34     B_modified <= B when sel = "00" else
35         not B when sel = "01" else
36             (others => '0');
37     Add_Cin <= '1' when sel = "01" else '0';
38     -- RCA performs addition or subtraction
39     RCA1: rca_16bit
40     port map (
41         A      => A,
42         B      => B_modified,
43         Cin    => Add_Cin,
44         Sum    => Add_Result,
45         Cout   => Add_Cout
46     );
47
48     -- Select the ALU operation
49     process(A, B, sel, Add_Result, Add_Cout)
50     begin
51
52         case sel is
53             when "00" =>          -- ADD
54                 Result <= Add_Result;
55                 Cout   <= Add_Cout;
56
57             when "01" =>          -- SUB
58                 Result <= Add_Result;
59                 Cout   <= Add_Cout;
60
61             when "10" =>          -- AND
62                 Result <= A and B;
63                 Cout   <= '0';
64
65             when "11" =>          -- OR
66                 Result <= A or B;
67                 Cout   <= '0';
68
69             when others =>
70                 Result <= (others => '0');
71                 Cout   <= '0';
72
73         end case;
74
75     end process;
76
77 end Structural;
```

Program 4: VHDL Testbench Code for 16-bit ALU

```

G:/code/vhdl/tb_alu_16bit.vhd (/tb_alu_16bit) - Default
Ln#
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity tb_alu_16bit is
5  |
6  |
7  | architecture Behavioral of tb_alu_16bit is
8  |
9  | component alu_16bit
10 | |
11 | | Port (
12 | |     A      : in  STD_LOGIC_VECTOR(15 downto 0);
13 | |     B      : in  STD_LOGIC_VECTOR(15 downto 0);
14 | |     sel    : in  STD_LOGIC_VECTOR(1 downto 0);
15 | |     Result : out STD_LOGIC_VECTOR(15 downto 0);
16 | |     Cout   : out STD_LOGIC
17 | | );
18 | | end component;
19
20 | signal A      : STD_LOGIC_VECTOR(15 downto 0) := (others => '0');
21 | signal B      : STD_LOGIC_VECTOR(15 downto 0) := (others => '0');
22 | signal sel    : STD_LOGIC_VECTOR(1 downto 0) := "00";
23 | signal Result : STD_LOGIC_VECTOR(15 downto 0);
24 | signal Cout   : STD_LOGIC;
25
26 | begin
27
28 | UUT: alu_16bit
29 | | port map (
30 | |     A      => A,
31 | |     B      => B,
32 | |     sel    => sel,
33 | |     Result => Result,
34 | |     Cout   => Cout
35 | | );
36
37 | stim_proc: process
38 | | begin
39 | |
40 | |     -- Test 1: ADD
41 | |     -- 5 + 3 = 8
42 | |     A <= x"0005";
43 | |     B <= x"0003";
44 | |     sel <= "00";
45 | |     wait for 10 ns;
46 | |
47 | |     -- Test 2: SUB
48 | |     -- 5 - 3 = 2
49 | |     A <= x"0005";
50 | |     B <= x"0003";
51 | |     sel <= "01";
52 | |     wait for 10 ns;
53 | |
54 | |     -- Test 3: AND
55 | |     -- 000A AND 000C = 0008
56 | |     A <= x"000A";
57 | |     B <= x"000C";
58 | |     sel <= "10";
59 | |     wait for 10 ns;
60 | |
61 | |     -- Test 4: OR
62 | |     -- 000A OR 000C = 000E
63 | |     A <= x"000A";
64 | |     B <= x"000C";
65 | |     sel <= "11";
66 | |     wait for 10 ns;
67 | |
68 | |     -- Test 5: ADD with carry
69 | |     -- FFFF + 0001 = 0000, Cout = 1
70 | |     A <= x"FFFF";
71 | |     B <= x"0001";
72 | |     sel <= "00";
73 | |     wait for 10 ns;
74 | |
75 | |     -- Test 6: SUB with borrow
76 | |     -- 3 - 5 = FFFE (16-bit 2's complement)
77 | |     A <= x"0003";
78 | |     B <= x"0005";
79 | |     sel <= "01";
80 | |     wait for 10 ns;
81 | |
82 | |     -- End simulation
83 | |     wait;
84 | | end process;
85
86 | end Behavioral;
87

```

Testbench Output: 16-bit ALU Simulation Results

